THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2014: Multivariable Calculus and Linear Algebra

December 16, 2021 TIME: 2:30 p.m.-5:00 p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL TEN questions. Each question carries 10 marks.

Notes:

- You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**
- You must start each question on a new page. You should write down the question number on the top right hand corner of each page.
- If your answer to a question spans over more than one page, you must indicate clearly on each page (except the last) that the answer will continue on the subsequent page.
- Whenever there is a blank page it will be assumed that all subsequent pages are also blank. If this is not true, indicate clearly on the page which you would otherwise leave blank.

- Some formulas:

$$
\begin{aligned}
\sin 2\theta &= 2\sin\theta\cos\theta \\
\cos 2\theta &= \cos^2\theta - \sin^2\theta \\
\text{For } n \geq 2&\text{:} \\
\int \cos^n\theta \, d\theta &= \frac{1}{n}\cos^{n-1}\theta\sin\theta + \frac{n-1}{n}\int \cos^{n-2}\theta \, d\theta \\
\int \sin^n\theta \, d\theta &= -\frac{1}{n}\sin^{n-1}\theta\cos\theta + \frac{n-1}{n}\int \sin^{n-2}\theta \, d\theta \\
\sin(\alpha+\beta) &= \sin\alpha\cos\beta + \cos\alpha\sin\beta \\
\sin(\alpha-\beta) &= \sin\alpha\cos\beta - \cos\alpha\sin\beta \\
\cos(\alpha+\beta) &= \cos\alpha\cos\beta - \sin\alpha\sin\beta \\
\cos(\alpha-\beta) &= \cos\alpha\cos\beta + \sin\alpha\sin\beta
\end{aligned}
$$

1. A semi-circular metal plate has varying thickness. The plate is given by $\{(x,y)|x^2+y^2 \leq 4 \text{ and } x \geq 0\}$ and the thickness at the point (x,y) is given by $H(x,y) = x + 2y + 10$. Find the thickest and thinnest points of the plate and the thickness at these points.

2. Find all critical points of the function $w = x + y + 4\sin x \sin y$. Also determine the nature of the critical points by second order derivative test (relative maximum, relative minimum or saddle point).

3. Using ONE iteration of Newton- Raphson's method, find the first approximate for the solution to the system of equations

$$\begin{cases} x^x + y^2 = 4 \\ y + \ln x = 1 \end{cases}$$

with initial guess $(2,0)$. (Round your answers to 4 decimal places.)

4. Find the volume of material cut from $x^2+y^2+z^2 \leq 4$ by the cylinder $x^2+y^2 = 2y$.

5. Evaluate the double integral $\int\int_R x^2 - xy + y^2 dA$, where R is the ellipse given by $x^2 - xy + y^2 \leq 2$. (Hint: you may use the transformation $x = \sqrt{2}u - \sqrt{\frac{2}{3}}v$, $y = \sqrt{2}u + \sqrt{\frac{2}{3}}v$.)

6. Find the center of mass of the lamina of density $\rho(x,y) = x + y$ occupying the region R under the curve $y = x^2$ in the interval $0 \leq x \leq 2$.

7. Let E be the region bounded below by the cone $z = \sqrt{x^2+y^2}$ and above by the sphere $z = x^2 + y^2 + z^2$.

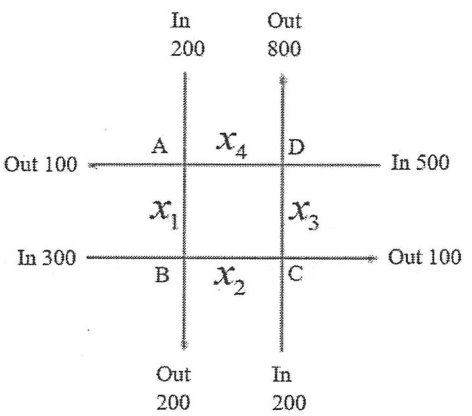
   (a) Set up a triple integral in spherical coordinates to find the volume of the region using the following orders of integration:

      (i) $d\rho d\phi d\theta$;

      (ii) $d\phi d\rho d\theta$;

   (b) find the volume of the region by using the integral in either (a)(i) or (a)(ii).

8. In the analysis of traffic flow, a certain city estimates the following situation for the "square" of its downtown district. In the following figure, the arrows indicate the flow of traffic. If x_1 represents the number of cars travelling between intersections A and B, x_2 represents the number of cars travelling between B and C, x_3 represents the number between C and D, and x_4 represents the number between D and A, we can formulate equations based on the principle that the number of vehicles entering an intersection equals the number leaving it. That is, for intersection A we obtain $200 + x_4 = 100 + x_1$. Formulate equations for the traffic at B, C, and D and solve the system of these four equations by Gaussian elimination.



9. Let $A = \begin{bmatrix} 5 & 0 & -2 \\ 2 & 3 & -2 \\ 0 & 0 & 3 \end{bmatrix}$.

   (a) Find all the eigenvalues and associated eigenvectors of A (Let the first non-zero entry of each eigenvector to be 1).

   (b) Find a non-singular matrix P such that $P^{-1}AP$ is a diagonal matrix.

   (c) Hence, evaluate A^{2022}.

10. Let V be an $n-$dimensional vector space over a field K. Suppose that $v_1, v_2, \ldots, v_k$ are linearly independent vectors in V. Determine whether the vectors $v_1 + v_2, v_2 + v_3, \ldots, v_{k-1} + v_k, v_k + v_1$ are linearly dependent or independent and prove your result.

******* **END OF PAPER** *******